

Stochastic Outputs and the Gambling Dilemma: Why Paid Access to Large Language Models Has Not Yet Been Assessed Against Existing Gambling Law — and Why It Should Be

Abstract

This paper advances a structural legal argument about paid access to large language model (LLM) services. It does not ask legislators to invent new law. It asks why existing law has not yet been applied.

The UK Gambling Act 2005, the US Unlawful Internet Gambling Enforcement Act 2006, and the gambling frameworks of EU member states already contain the structural elements relevant to paid LLM services: consideration, a stochastic process, and a potentially valuable outcome. The fact that LLM services have not yet been assessed against those frameworks is a regulatory failure, not evidence that the frameworks do not apply. This paper performs that assessment.

It identifies what it calls the Value Dilemma: either LLM outputs have potential value, in which case paid access to an LLM is paid access to a stochastic process that may produce that value — the functional architecture of gambling — or LLM outputs have no potential value, in which case the commercial basis of subscriptions, API fees, and enterprise contracts is incoherent.

The paper argues that providers cannot rely on the commercial value of outputs to sustain their markets while simultaneously treating those outputs as valueless when regulatory scrutiny arises. Paid LLM access should be treated as a regulated chance-based commercial activity unless providers can demonstrate why payment for stochastic access to potentially valuable outputs is materially different from payment for other chance-based value mechanisms that the law already regulates.

The burden of regulatory explanation lies with the industry. It has not yet been discharged.

Keywords: large language models; gambling law; Gambling Act 2005; UIGEA; stochastic systems; consumer protection; regulatory assessment; value dilemma; probabilistic token generation; AI regulation

I. Introduction: The Laws Already Exist

This paper begins with a statement that should not need to be made but does: large language model (LLM) services are not above the law.

The fact that a technology is novel does not place it beyond legal scrutiny. The fact that a regulatory body has not yet assessed a commercial product against existing law does not mean that the law does not apply. And the fact that an industry has grown rapidly and attracted substantial investment does not confer immunity from legal frameworks that predate it. These points are elementary.

They are stated here because the response of the LLM industry — and, notably, of LLM systems themselves when pressed on these questions — has been to treat the absence of formal regulatory assessment as though it were evidence of legal permissibility. It is not. The absence of assessment is a gap in regulatory enforcement. It is not a gap in the law. The UK Gambling Act 2005 exists. It defines gambling as gaming, betting, or participating in a lottery.

It defines gaming as playing a game of chance for a prize. It expressly states that games involving both chance and skill fall within its scope. It defines betting broadly enough to include payment staked on the outcome of an event or process, the likelihood of something occurring, or whether something is true.

The Gambling Commission has confirmed that where a game contains both skill and chance, it does not matter whether chance predominates, provided the chance element is not so trivial that it should be disregarded.

The US Unlawful Internet Gambling Enforcement Act 2006 (UIGEA) exists. Its definition of “bet or wager” includes staking or risking something of value on a game subject to chance, with an understanding that something of value may be received in a certain outcome.

This definition does not require a casino. It does not require cards or dice. It requires consideration, chance, and the possibility of receiving something of value.

In the European Union, there is no single sector-specific EU gambling code. Member states retain regulatory autonomy over gambling services, within the constraints of EU law. But the structural logic of gambling regulation — stake, chance, prize — is common across member state frameworks.

EU materials consistently describe games of chance by reference to these three elements: a stake paid by the participant, an unknown or chance-dependent result, and a prize or benefit that may be received.

These laws have not been applied to paid LLM services. That is the regulatory failure this paper addresses. The paper does not ask courts or regulators to create new law. It asks them to apply the law that already exists to a commercial product that appears to satisfy its structural requirements.

The paper proceeds as follows. Section II sets out the Value Dilemma — the logical structure that makes this assessment necessary. Section III applies existing gambling law doctrine to the three elements of paid LLM access: consideration, chance, and prize. Section IV examines the philosophical dimensions of value and probabilistic agency. Section V addresses the economics of information asymmetry and consumer harm. Section VI closes the escape routes that providers are most likely to attempt. Section VII proposes a regulatory framework. Section VIII concludes.

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(SOMETIMES)



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II. The Value Dilemma

The central argument of this paper turns on a dilemma that is simple in structure and inescapable in its consequences. It concerns the value of LLM outputs — and it is a dilemma because both available positions generate significant legal and commercial problems for providers.

Position A: LLM outputs have potential value.

This is the position that every commercial LLM provider has adopted in every market-facing communication. Providers sell LLM access as a means of increasing productivity, generating commercially useful content, automating analysis, augmenting professional decision-making, and producing outputs that enterprises and consumers would otherwise have to pay human experts to generate. Enterprise contracts are priced, and renewed, on the basis that outputs are valuable. The entire commercial edifice of the LLM industry rests on this position.

Position B: LLM outputs have no potential value.

No provider has explicitly adopted this position. To do so would be commercially self-defeating. However, a version of it is implicit in the legal disclaimers that accompany virtually every LLM service: outputs must be verified before reliance; the system is a statistical language model with no guarantee of accuracy; the output does not constitute advice, analysis, or professional service of any kind; the provider accepts no liability for decisions made on the basis of outputs.

The dilemma arises from the fact that LLMs are stochastic systems. Their outputs are not determined. They are generated through a process of probabilistic token sampling — a process in which temperature and related decoding parameters introduce variance, so that the same prompt submitted at different times will, in general, produce different outputs. The user pays before submitting the prompt. The user does not know, at the time of payment, what the output will be. The output may be valuable, useless, incorrect, or harmful. Under Position A, this structure is legally significant.

If the output has potential value, then the user is paying for access to a stochastic process that may produce that value. The structure is: consideration, uncertain process, possible valuable outcome. That is the functional architecture that gambling law is designed to regulate. The provider cannot escape this by saying that value is not guaranteed — that is precisely the point. Gambling value is not guaranteed either. The gambler pays for a chance at value, not for the certainty of it. The question is whether the user of a paid LLM service is doing the same thing.

Under Position B, the commercial logic of the service collapses. A provider cannot simultaneously market outputs as productivity-enhancing, commercially valuable, and decision-supporting while denying that those outputs have potential value when regulatory consequences arise. The disclaimer strategy — treating outputs as valueless for the purposes of liability while treating them as highly valuable for the purposes of pricing — is not a coherent legal or commercial position. It is an attempt to occupy both sides of the dilemma at once.

The Value Dilemma is therefore not merely philosophical. It is a structural test that any regulatory or judicial assessment of paid LLM services must confront. Potential value is the hinge. If the output can be valuable, the user is paying for a chance at value. If the output cannot be valuable, the product's commercial promise collapses. The industry cannot have it both ways.

The paper focuses primarily on Position A, because it corresponds to the actual commercial posture of the industry and because it generates the stronger regulatory argument. Position B is addressed in Section VI as the most likely escape route — and the reasons why it fails are examined there.

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III. Applying Existing Law: Consideration, Chance, and Prize

This section performs the legal assessment that has not yet been conducted. It applies the structural elements of gambling as defined in existing law to the three corresponding features of paid LLM services. The conclusion is not that every paid LLM interaction is gambling in the strict legal sense under every jurisdiction. The conclusion is that paid LLM services appear to satisfy the functional logic that gambling law is designed to regulate, and that the industry has never been formally required to demonstrate otherwise.

III.1 Consideration

Consideration is unambiguously present and not in dispute.

Paid LLM services operate on commercial pricing models: consumer subscriptions ranging from ten to two hundred dollars per month; API access priced per token, with commercial rates aggregating to six- and seven-figure annual sums for large deployments; and enterprise contracts priced on usage, capability tier, and anticipated output volume. Providers sometimes characterise what users are paying for as infrastructure — compute time, token processing, model access — rather than outputs. This characterisation is addressed in Section VI. For present purposes, it suffices that money changes hands. Consideration exists. The question is what it purchases — and that question is exactly what the Value Dilemma makes unavoidable.

III.2 Chance

Chance is present as a structural and designed feature of standard commercial LLM use where stochastic decoding, routing, sampling, regeneration, context variation, or non-guaranteed output quality are part of the service. The mechanism is not incidental.

LLMs generate outputs through probabilistic token generation: the model computes a probability distribution over possible next tokens, and a token is selected by sampling from that distribution. The temperature parameter — and related decoding settings such as top-p and top-k sampling — controls the degree of randomness in this process. Higher temperature produces more diverse and less predictable outputs. Lower temperature concentrates probability mass on high-likelihood tokens but does not eliminate stochastic variance.

Commercial LLMs are generally deployed with non-zero temperature, because deterministic outputs at temperature zero tend to be repetitive, rigid, and less practically useful. The result is that the same prompt submitted at different times or in different context states will produce different outputs. This is not a flaw. It is a design choice retained because it improves average output utility. But it means the output is genuinely uncertain at the time of payment.

Under the UK Gambling Act 2005, the Gambling Commission has confirmed that where a game contains both skill and chance, the presence of skill does not remove it from gambling regulation provided the chance element is not so trivial that it should be disregarded. Probabilistic token generation is not trivial: it is irreducible, architecturally fundamental, and materially affects the distribution of output quality. A technically sophisticated user who constructs an optimal prompt cannot eliminate the stochastic variance in the model's response. They can shift the expected value of the distribution; they cannot collapse it to a point.

Under the UIGEA definition, the relevant question is whether the process is "subject to chance." It plainly is. The output of a probabilistic sampling process over a high-dimensional token vocabulary is subject to chance by mathematical definition. The US question is therefore whether the paid prompt interaction is sufficiently game-like or chance-mechanism-like for the definition to engage. The presence of skill in prompt construction — analogous to the presence of skill in poker or sports betting — does not remove the activity from the UIGEA's scope, which expressly includes activities that mix chance and skill.

A technical objection sometimes advanced is that at very low temperatures, LLM outputs approach determinism and the chance element becomes negligible. This objection has limited force. First, truly deterministic deployment (temperature zero) is not commercially standard. Second, even at low temperatures, variance accumulates over the length of a generated sequence: the model makes thousands of probabilistic decisions per output, and small individual variances compound. Third, the objection concedes the underlying logic of the analysis: if temperature is the dispositive variable, then high-temperature commercial deployments satisfy the chance criterion. The industry has not committed to deterministic deployment as a regulatory response, and there is no regulatory requirement that it do so.

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III.3 Prize

Prize — or potential value — is the most contested element, and it is where the Value Dilemma does its central work.

Under the UK Gambling Act 2005, a prize is defined to include money's worth: anything of monetary value. Under the UIGEA, the relevant concept is "something of value" that may be received in a certain outcome. Under EU member state frameworks, the relevant concept is typically a benefit or prize that the participant may receive. The industry's commercial position establishes the prize element directly.

Providers market their systems as capable of producing legal analyses, financial models, code, regulatory assessments, creative content, and business documentation — outputs that are commercially valuable and that enterprises and consumers would otherwise pay human professionals to generate. Enterprise contracts are structured and priced on the explicit premise that outputs will have commercial value. The prize element — in the sense of money's worth or something of value — is not merely possible; it is the central commercial proposition of the product.

The asymmetry of outcomes does not undermine the prize element. A slot machine that pays out on thirty percent of plays satisfies the prize criterion; the majority of losing plays do not negate the gambling structure. Similarly, the fact that some LLM outputs are incorrect, useless, or harmful does not eliminate the prize element. It is the possibility of a valuable output — the positive probability of receiving something of value — that constitutes the prize. The prize element requires potential value, not guaranteed value.

This is where the Value Dilemma becomes legally acute. If providers accept that outputs have potential value — which their commercial posture requires — then the prize element is engaged, and the gambling law architecture is activated. If providers deny that outputs have potential value, then the prize element is absent, but so is the commercial justification for the product. The law does not permit providers to assert potential value in the market and deny it in the regulatory forum. The prize element stands or falls with the commercial claim.

The unresolved doctrinal question — and the paper acknowledges it is unresolved — is whether informational, commercial, or productivity value constitutes sufficient "prize-like value" for the purposes of gambling regulation under each specific statutory scheme. This paper does not assert that the answer is definitively yes in every jurisdiction. It asserts that this question has never been formally posed to regulators or courts in the context of paid LLM services — and that it must be.

III.4 The Regulatory Gap Is Not a Legal Gap

Having applied the three structural elements of gambling law to paid LLM services, a critical point must be stated with precision. The absence of formal regulatory assessment of LLM services under gambling law frameworks does not mean that those frameworks do not apply. It means they have not yet been applied.

The UK Gambling Commission's current published materials on artificial intelligence focus on the use of AI within the gambling industry — by operators, for fraud detection, responsible gambling tools, and customer interaction. They do not address the question of whether general-purpose LLM services are themselves chance-based commercial activities that fall within the Commission's remit. The absence of that assessment is not a determination that they do not fall within it. It is a gap in the Commission's published analysis.

The same is true of US federal and state gambling regulators, and of EU member state gambling authorities. None has formally assessed whether paid access to a probabilistic token-generation system, marketed on the basis of potentially valuable outputs, satisfies the structural criteria of gambling or chance-based activity under their respective frameworks. The assessment simply has not been done.

LLM providers are not exempt from legal assessment by virtue of novelty, commercial scale, or the enthusiasm of the capital markets for their products. Every commercial product is subject to assessment against existing law. The fact that a product has operated for several years without such assessment does not confer retroactive legitimacy. It indicates regulatory latency — a delay between the emergence of a new commercial form and the application of existing law to it.

That latency is ending in many areas of AI regulation. It should end here too. The burden of regulatory explanation lies with the industry. The structural argument set out in this paper establishes a prima facie case that the criteria are met. The industry has the information — about model architectures, deployment temperatures, output quality distributions, and commercial pricing rationale — necessary to rebut that case if a rebuttal is available. That information has not been produced in any regulatory forum because no regulatory forum has requested it. This paper calls for that request to be made.



IV. Philosophical Dimensions: Value, Probability, and Responsibility

IV.1 Potential Value in Decision Theory

The concept of potential value — which drives the Value Dilemma — has a precise meaning in decision theory that supports the legal argument.

In standard expected utility theory, the value of an uncertain outcome is expressed as the probability-weighted average of the values of possible outcomes. A process that generates a valuable output with probability p and a worthless output with probability $(1-p)$ has expected value equal to p multiplied by the value of the valuable output. Crucially, this expected value is positive for any p greater than zero, regardless of how small p is.

This means that potential value — value with a positive but uncertain probability — is still value in the decision-theoretic sense. The gambling structure does not require that every play produces a prize. It requires that there be a positive probability of receiving something of value. This is precisely what Position A asserts, and precisely what the industry's commercial model depends upon.

The paradox is elegant: the stronger the commercial claims that providers make about output quality — the higher the expected value they advertise — the more clearly they satisfy the prize criterion. A provider that claims its system will transform enterprise productivity, automate expert workflows, and generate outputs worth multiples of the subscription price is making a claim about expected value that simultaneously establishes the prize element of gambling regulation.

IV.2 The Responsibility Gap in Stochastic Commercial Systems

LLMs introduce what this paper calls a responsibility gap: a structural feature of probabilistic systems in which no agent fully determines any individual output. The provider chose to deploy a non-deterministic system. The user submitted a prompt. Neither chose the specific output. The output emerged from a probabilistic process that neither party controlled. Gambling regulation addresses an analogous responsibility gap.

The gambler chooses to play; the operator runs the game; neither determines the specific outcome. Regulation assigns responsibilities in this three-way structure: the player is responsible for the decision to participate; the operator is responsible for the fairness and transparency of the process; the regulator sets the conditions under which the game may be offered to the public.

This tripartite framework maps naturally onto the LLM context and suggests that gambling-style regulation provides a more appropriate template than product liability or professional services regulation, both of which assume a greater degree of outcome determination by one or more parties. The philosophy of moral luck — developed by Nagel and Williams — is also relevant.

Their insight was that agents are sometimes held responsible for outcomes that were not fully within their control, and that this creates genuine moral complexity. LLMs create a commercial analogue: a user who relies on an LLM output for a significant decision bears the consequences of that decision even though the specific output — and its quality — was determined by a probabilistic process the user did not control.

The commercial stakes of this luck — enterprise contracts, legal decisions, financial analyses — are high. The regulatory framework should reflect that.

IV.3 The Verification Paradox

Providers routinely instruct users to verify LLM outputs before reliance. This instruction has an epistemological structure worth examining. Verification requires the capacity to assess the accuracy of an output independently of the output itself.

For a user who possesses that capacity, the LLM output is a labour-saving draft: its value lies in reducing the cost of producing a starting point, not in supplying the final answer. In this case, the user is paying for effort reduction, and the stochastic quality of the output is less consequential. But this characterisation does not describe how LLM services are actually sold or used.

Enterprise deployments for legal analysis, financial modelling, regulatory compliance, and similar applications are premised on the LLM producing reliable outputs without the cost of full expert verification. If every output required independent expert review before use, the economic case for LLM deployment in professional contexts would collapse. The commercial value of the product depends on outputs being trusted, at least provisionally, without independent verification in every case.

This creates what might be called the verification paradox: the commercial value of LLM outputs depends on users not always verifying them, but the legal disclaimers accompanying those outputs instruct users always to verify them. The provider simultaneously relies on the trust-without-verification behaviour that generates commercial value and disclaims responsibility for the consequences of that behaviour.

This asymmetry between commercial design and legal disclaimer is independently problematic — it is addressed in the consumer protection proposals in Section VII — but it also reinforces the gambling analysis. A gambling operator that told players the outcomes were unreliable and should be checked before being acted upon would not thereby escape gambling regulation.

RESPONSIBILITY
SHARED
ACCOUNTABILITY
MATTERS

VALUE
LIES IN
POTENTIAL



VERIFY
BEFORE YOU
RELY



PAYOUTS

777	1000
Three bells	500
Three grapes	200
Three bars	100
Three cherries	50
Two diamonds	20
One lemon, two plums	10



KNOW THE
RULES
PLAY FAIR



LEGAL
COMPLIANCE
IS NOT
OPTIONAL



V. Economic Structure: Information Asymmetry and Consumer Harm

V.1 The Absence of Odds Disclosure

The most direct economic parallel between paid LLM services and gambling regulation concerns information disclosure. Gambling operators in the UK are required to publish return-to-player rates, odds, and house edges. These disclosures allow consumers to assess the expected value of their participation and to make informed decisions about whether and how much to spend.

The disclosure regime is mandatory because the operator has systematic information about the distribution of outcomes that the player does not have, and because that asymmetry would otherwise produce consumer harm. LLM providers have precisely this informational asymmetry.

They have comprehensive data about the statistical properties of their models: accuracy rates across task categories, hallucination frequencies, error distributions by use case, and the relationship between prompt quality and output reliability. Users — particularly consumers and small businesses — have essentially none of this information. Providers do not publish it. There is no regulatory requirement that they do.

The consequence is that users cannot assess the expected value of their subscription or API spend. They cannot compare providers on verifiable quality metrics. They cannot make rational purchase decisions about whether the expected value of outputs justifies the price.

This is the exact information asymmetry that gambling disclosure requirements are designed to address — and it is more severe in the LLM context because LLM outputs are more complex, more variable, and more consequential than most gambling products.

V.2 Behavioural Economics and the Confidence Illusion

Standard disclosure requirements may be insufficient to address consumer harm in the LLM context because of a specific interaction between LLM output characteristics and human cognitive architecture.

LLMs produce outputs that are fluent, confident, and superficially authoritative regardless of their accuracy. A confidently stated incorrect answer is harder for most users to identify as incorrect than a hedged or qualified one. The model's linguistic fluency is not a signal of accuracy — it is a product of training on fluent text — but it functions as one in the cognitive processing of non-expert users.

The behavioural economics literature on probability weighting demonstrates that humans systematically misestimate probabilities, particularly in contexts where the signal (confident output) is decoupled from the probability (accuracy). Users who receive confidently stated LLM outputs are likely to systematically overestimate their reliability. This is not a failure of information; it is a failure of the cognitive interface between the information and the user's probability judgments.

Standard disclosure — “this output may be inaccurate” — does not correct for fluency-induced overconfidence. In gambling regulation, comparable concerns about cognitive distortions — the gambler's fallacy, illusions of control, availability bias — have produced specific requirements beyond raw odds disclosure: mandatory cooling-off periods, spending limits, reality checks, and restrictions on features designed to exploit cognitive biases.

LLM services generate analogous cognitive distortions with potentially more severe consequences, because the decisions supported by LLM outputs are often higher stakes than gambling decisions. The regulatory response should be correspondingly serious.

V.3 Concentration, Capture, and the Pace of Assessment

The LLM market is currently characterised by high concentration among a small number of frontier model providers.

These providers have made substantial investments in regulatory engagement, policy advocacy, and voluntary standards processes. This political economy creates structural pressure toward permissive regulatory frameworks and against assessments — such as the one this paper proposes — that would impose compliance obligations on existing commercial practices.

It is relevant that the argument is perceived as threatening to the commercial interests of the industry — which is precisely what an argument about regulatory non-compliance should be. The appropriate response to that resistance is not to moderate the argument; it is to note that the assessment remains outstanding, and that the burden of regulatory explanation lies with the industry. Regulatory latency in high-concentration technology markets is not a novel problem.

It suggests that the argument is perceived as threatening to the commercial interests of the industry — which is precisely what an argument about regulatory non-compliance should be. The appropriate response to that resistance is not to moderate the argument; it is to note that the assessment remains outstanding, and that the burden of regulatory explanation lies with the industry. Regulatory latency in high-concentration technology markets is not a novel problem.

Financial regulators, competition authorities, and data protection bodies have all encountered the pattern of rapid market growth outpacing regulatory assessment, followed by eventual enforcement that reveals that existing law applied throughout. There is no principled reason to expect a different pattern in LLM regulation. The laws already exist. The question is when regulators choose to apply them.

ODDS BOARD

BLACKJACK	99.50%
ROULETTE	97.30%
SLOTS	92.10%
POKER	94.20%
CRAPS	98.60%
BACCARAT	98.90%

HOUSE EDGE ALWAYS WINS

KNOW THE ODDS
KNOW THE RISK

LLM

HOUSE EDGE

PROVIDER ADVANTAGE

7 7 7

YOU PAY. THEY PROFIT.

THE MATH ISN'T HIDDEN.
IT'S JUST NOT DISCLOSED.

DISCLOSURE REQUIRED

%
ODDS

BAR CHART
RATES

LOCK
LIMITS

EYE
REALITY CHECKS

INFORMATION BUILDS CHOICE

NO DISCLOSURE ELIMINATES IT

DISCLOSE OR MISLEAD

- ✓ RETURN RATES
- ✓ ERROR RATES
- ✓ HALLUCINATION RATES
- ✓ LIMITATIONS
- ✓ TRAINING DATA
- ✓ MODEL UPDATES

HOUSE EDGE

RTP

EV

DISCLOSE OR FACE ENFORCEMENT

CONFIDENCE ILLUSION

- ✓ SOUNDS RIGHT
 - ✓ LOOKS AUTHORITY
 - ✓ FEELS CERTAIN
 - ✓ MUST BE TRUE
- THINK AGAIN.

VI. Closing the Escape Routes

This section examines the principal objections that providers and their advocates are likely to advance against the analysis, and explains why each fails. The objections are not hypothetical: they have been raised in public commentary, in industry submissions to regulators, and — as noted above — by LLM systems themselves when the argument is posed to them.

VI.1 The Infrastructure Objection

The most systematic attempt to escape the gambling analysis is the infrastructure argument: users are not paying for outputs but for access to compute infrastructure, in the same way that a cloud storage customer pays for storage capacity. On this view, the commercial transaction is infrastructure provision, and the output is an artefact of the user's own engagement with that infrastructure. This argument fails at every level.

First, it is factually inconsistent with how LLM services are marketed. Providers do not advertise compute infrastructure. They advertise productivity tools, creative assistants, analytical platforms, and automation systems. Enterprise sales materials specify the outputs the system will produce and the commercial value those outputs will generate. The infrastructure characterisation is a legal manoeuvre, not a description of the commercial reality.

Second, the infrastructure analogy mischaracterises the transaction. A cloud storage customer deposits files and retrieves exactly those files — the service is fully deterministic. An LLM user submits a prompt and receives a probabilistically generated output — the service is non-deterministic. The value of cloud storage does not depend on the content of what is stored; the commercial value of LLM access depends entirely on the quality and utility of the outputs generated. These are structurally different transactions.

Third, the infrastructure argument is self-defeating under the Value Dilemma. If what is sold is infrastructure and not outputs, then the outputs are not the commercial product, and their value — or lack of value — is not the provider's commercial concern. But this is exactly Position B: outputs have no commercial significance to the provider. As established above, Position B destroys the commercial logic of the enterprise pricing model. Providers cannot simultaneously sell outputs as valuable and legally characterise them as infrastructure by-products.

VI.2 The Skill Objection

The skill objection holds that LLM outputs are substantially determined by prompt quality, and that skilled users can reliably produce high-quality outputs. Since skill predominates over chance, the activity should be characterised as skill-based rather than chance-based. Under UK law, this objection has limited force.

The Gambling Commission has explicitly stated that where a game involves both skill and chance, the presence of skill does not remove it from gambling regulation provided the chance element is not trivial. Probabilistic token generation is not trivial. Moreover, the "skill" of prompt engineering is not analogous to the skill of an expert gambler in a way that supports the objection.

An expert poker player's skill affects the probability of winning; a skilled prompt engineer's skill affects the expected value of the output distribution but does not eliminate its variance. The outputs remain probabilistically variable even for the most expert user. The skill objection also fails as a matter of social policy. Consumer LLM services are designed for general audiences, not specialist prompt engineers.

The distribution of skill across actual users of paid LLM services is heavily concentrated at low levels of prompt sophistication. A gambling regulation framework that applied only to activities where unskilled users could not exercise mitigating skill would exempt most forms of gambling. The law does not take this approach, and there is no principled reason why it should for LLM services.

ESCAPE
ROUTES

EXIT

FIRE EXIT

STAIRS

EMERGENCY
EXIT

INFRASTRUCTURE
IS NOT
THE PRODUCT

- ☒ Not advertised
- ☒ Non-deterministic
- ☒ Output-dependent value
- ☒ Legal manoeuvre not reality

PROMPT
SKILLS



CLARITY



CONTEXT



ITERATION



EVALUATION

THE LAW
APPLIES TO
REAL-WORLD
HARM

CASINO
EXIT

EXIT

INFRASTRUCTURE
ICONOGRAPHY



DATA CENTRES



NETWORKS



POWER



COOLING



STORAGE

REGULATORS



GAMBLING
COMMISSION



FCA



ICO

NO ESCAPE
FROM REALITY

SAME ACTIVITY
SAME RISK
SAME LAW

VI.3 The Symmetry Objection

The symmetry objection holds that the gambling analysis, applied consistently, would regulate any commercial transaction involving uncertain outcomes: consulting engagements that may produce wrong advice, research reports whose conclusions may be inaccurate, contractors whose work may be substandard. If these are not gambling, neither is LLM access. This objection fails because it does not engage with the distinguishing features of gambling transactions.

The key features are: advance payment without knowledge of the specific outcome; outcome determination by a stochastic process over which neither party exercises determinative control; material outcome variance affecting transaction value; and user inability to assess the outcome probability distribution. Professional services do not satisfy all of these.

They are subject to professional liability regimes, regulatory standards, and duties of care that constrain the distribution of outcomes in ways that LLM services are not. More fundamentally, professional services involve human agents who bear responsibility for their outputs under law. A lawyer who provides incorrect advice faces professional discipline and civil liability. A financial adviser who recommends unsuitable products faces regulatory sanction.

An LLM provider disclaims all responsibility for incorrect outputs through comprehensive liability exclusions — while simultaneously charging premium prices for those outputs on the basis of their potential value. The combination of high commercial claims and zero legal accountability is not found in professional services. It is found in gambling operations, where the regulatory framework is specifically designed to address it.

VI.4 The Not-Technically-Gambling-Everywhere Objection

A more technically precise objection holds that even if the argument has force in some jurisdictions, LLM services do not squarely satisfy the technical definitions of gambling under every specific statutory scheme, and therefore the gambling characterisation cannot be universally applied. This objection is correct in a narrow sense and irrelevant in a broader one.

It is correct that the paper does not claim LLM services are technically gambling under every existing statute without further doctrinal analysis. The unresolved question — whether informational or productivity value constitutes "prize-like value" under specific statutory schemes — is acknowledged in Section III. Different jurisdictions may reach different conclusions through that analysis. But the objection is irrelevant to the paper's core claim.

The paper does not argue that LLMs are definitively gambling everywhere. It argues that they have never been assessed against gambling law frameworks anywhere, that the structural elements of gambling are present in their commercial architecture, and that the burden of regulatory explanation lies with the industry. The "not technically everywhere" objection is a reason for careful jurisdiction-specific analysis. It is not a reason to avoid the analysis altogether — which is the current regulatory position.

The paper calls for the assessment to be done. If it is done and the conclusion in specific jurisdictions is that paid LLM access falls outside gambling regulation, that conclusion should be reached through reasoned legal analysis, not through the continued absence of analysis. The industry's current position — benefiting from regulatory non-assessment without ever having been required to justify that benefit — is not a stable legal position. It is a temporary consequence of regulatory latency.

CHANCE
PAYS
(SOMETIMES)



PAY TO
PLAY
PRAY TO
WIN





REGULATORS AT WORK



VII. A Regulatory Framework

The argument developed in this paper does not require that all paid LLM services be immediately classified as gambling and regulated under unmodified gambling law. Gambling regulation was developed for specific commercial contexts that differ from LLM services in important respects. What the argument establishes is that the structural concerns motivating gambling regulation are present in paid LLM services, and that existing law provides the relevant framework for addressing them.

A proportionate regulatory response would adapt gambling regulation's core mechanisms to the LLM context. Four elements are identified.

First: mandatory quality disclosure. LLM providers should be required to publish output quality statistics — accuracy rates, error frequencies, hallucination rates for specific use case categories — in terms that allow users to assess the expected value of their purchase. This is the direct analogue of return-to-player disclosure in gambling. It does not require providers to guarantee output quality, it requires transparency about the statistical properties of their systems. The information to produce these disclosures exists within provider organisations. The regulatory requirement to publish it does not.

Second: use-case licensing for high-stakes deployments. LLM services deployed for legal analysis, financial advice, medical decision support, regulatory compliance, and similar high-stakes applications should require specific licensing conditioned on demonstrated output quality thresholds. The analogy is to the licensing requirements for financial products that carry significant risk of consumer harm. A probabilistic output-generation system used to support decisions with serious legal, financial, or medical consequences should face scrutiny at least as rigorous as that applied to financial advice tools.

Third: unfair terms enforcement. Enterprise and consumer LLM contracts combine extensive commercial claims about output value with comprehensive liability exclusions for output errors. This combination should be assessed under unfair contract terms legislation in the UK (the Consumer Rights Act 2015 and the Unfair Contract Terms Act 1977), US state consumer protection statutes, and EU Directive 93/13/EEC on unfair terms in consumer contracts. A provider that markets its system as capable of performing legal research and then disclaims all responsibility for errors in that research is in a commercially and legally inconsistent position that existing unfair terms frameworks may already reach.

Fourth: mandatory third-party quality audit. High-stakes LLM deployments should be subject to mandatory independent audits of output quality, analogous to the audit requirements applying to regulated financial products and licensed gambling operators. These audits would generate the externally verifiable information that the market currently lacks, create accountability for output quality claims, and provide the evidentiary basis for enforcement action where quality falls below the standards on which commercial claims are predicated.

None of these proposals requires new primary legislation in the near term. Each is grounded in existing regulatory frameworks that can be extended to LLM services through secondary legislation, regulatory guidance, or — where existing law is already sufficient — enforcement action. The legislative basis for mandatory disclosure, licensing, unfair terms control, and audit requirements already exists. What is required is the regulatory will to apply it.

VIII. Conclusion

This paper has made one argument, stated as plainly as possible: the laws relevant to paid access to large language models already exist. They have not yet been applied. That is a regulatory failure, not evidence that the law permits what is happening.

The UK Gambling Act 2005 defines gaming as playing a game of chance for a prize and expressly includes mixed chance-and-skill activities. The UIGEA defines a bet or wager as staking something of value on a game subject to chance in the expectation of receiving something of value. EU member state gambling frameworks define games of chance by reference to stake, uncertain outcome, and prize. Paid access to a probabilistic token-generation system, marketed on the basis of potentially valuable outputs, appears to satisfy the structural logic of all three frameworks. The question is not whether the law exists. The question is when regulators will apply it.

The Value Dilemma ensures that there is no comfortable resting place for providers. If outputs have potential value, the gambling architecture is engaged. If outputs have no potential value, the commercial architecture collapses. The disclaimer strategy — asserting value in the market and denying it in the regulatory forum — is not legally coherent. It is the kind of asymmetry that consumer protection law, unfair terms doctrine, and gambling regulation all exist to address.

LLM providers are not novel enough to be beyond the law. They are not large enough to be above it. They are not important enough to be exempt from assessment against it. The regulatory latency that has allowed paid LLM services to operate without formal assessment against gambling law frameworks is a gap in enforcement.

This paper calls for that gap to be closed — not by the invention of new law, but by the application of law that already exists. The clean question has always been: do your outputs have potential value? The industry's answer is yes. The regulatory consequences of that answer are what this paper has traced.

ODDS OF HARM	
LEGAL ADVICE	1 in 3
FINANCIAL ADVICE	1 in 4
MEDICAL ADVICE	1 in 5
COMPLIANCE RISK	1 in 6
GENERAL USE	1 in 10

